# Linear Algebraic Groups

Based on Lectures given by
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## 1 Introduction

The course concerns with algebraic groups. Primary examples are

**Example 1.1.** $\mathrm{GL}_2(k)$ *for any field* $k$.

**Remark 1.1.** *ALmost all finite simple groups arise as* $G(k)/Z_G(k)$ *for* $k$ *a finite field and* $G$ *a simple simply connected algebraic group.*

### 1.1 Recollection

**Definition 1.1.** *A bi-product* $(A, p_1 : A \to A_1, A \to A_2)$ *in a category* $\mathscr{C}$ *is an object which is universal in terms of any object* $B$ *together with morphisms* $f_1 : B \to A_1, f_2 : B \to A_2$. *That is, we have a unique morphism* $B \to A$. *We denote such* $A$ *to be* $A_1 \times A_2$.

*We define a finite product for* $n$*-many object similarly. For* $n = 0$, *such product is called the terminal object.*

**Notation 1.1.** *As the terminal object* $1_{\mathscr{C}}$ *admits a unique morphism into it from any object* $A$, *we denote such morphism as* $1_A : A \to 1_{\mathscr{C}}$.

**Remark 1.2.** *For* $A \in \mathscr{C}$, $A \times 1_{\mathscr{C}} \to A$ *given by the projection is an isomorphism with inverse* $\mathrm{id}_A \times 1_A$.

**Notation 1.2.** *For* $A \in \mathscr{C}$, $\Delta : A \to A \times A$ *is induced from* $A \overset{\mathrm{id}_A}{\to} A$.

**Definition 1.2.** *Let* $\mathscr{C}$ *be a category admitting finite products. A group object in* $\mathscr{C}$ *is a tuple* $(G, m, i, e)$ *such that*

$$G \in \mathrm{Ob}(\mathscr{C}), m : G \times G \to G, i : G \to G, e : 1_{\mathscr{C}} \to G,$$

*satisfying the following properties.*

1. $m$ *is associative. (i.e.* $m \circ (m \times \mathrm{id}_G) = m \circ (\mathrm{id}_G \times m)$*).*
2. $e$ *is a* 2*-sided unit. (i.e.* $m \circ (\mathrm{id}_G \times e) = \mathrm{id}_G = m \circ (e \times \mathrm{id}_G)$*).*
3. $i$ *is a* 2*-sided inverse. (i.e.* $m \circ (i \times \mathrm{id}_G) \circ \Delta = e \circ 1_G = m \circ (\mathrm{id}_G \times i) \circ \Delta$*).*

**Definition 1.3.** *A morphism of group objects is a morphism* $f : G_1 \to G_2$ *such that*

$$m_2 \circ (f \times f) = f \circ m_1, i_2 \circ f = f \circ i_1, e_2 = f \circ e_1.$$

*Objects of groups together with these morphism form a subcategory which we will denote by* $\mathrm{Grp}\mathscr{C}$.

**Remark 1.3.** *If* $G$ *is a gropu object in* $\mathscr{C}$, *then for any* $A \in \mathrm{Ob}(\mathscr{C})$, $\mathscr{C}(A, G)$ *inhertis a gropu structure as folows.*

1. $e_A := e \circ 1_A, A \to 1 \to G$,
2. $f_1 \cdot f_2 := m \circ (f_1 \times f_2) \circ \Delta_A$.
3. $f^{-1} := i \circ f$.

*Furthermore, the assingment $A \mapsto \mathscr{C}(A, G)$ is a contravariant functor $\mathscr{C} \to \mathrm{Grp}$. Then $G \mapsto \mathscr{C}(\cdot, G)$ is a contravariant functor $\mathrm{Grp}\mathscr{C} \to \mathrm{Func}(\mathscr{C}^{\mathrm{op}}, \mathrm{Grp})$ which is an embedding (i.e. fully-faithful) by Yoneda lemma.*

*This is called the functor of points, we always have the abstract test group $G(1)$. In general, we will denote $G(A) := \mathscr{C}(A, G)$ for arbitrary $A \in \mathrm{Ob}(\mathscr{C})$.*

**Example 1.2.** *In $\mathscr{C} = \mathrm{Sets}$, the terminal object is the singleton set and $\mathrm{Grp}\mathscr{C} = \mathrm{Grp}$. $G(1)$ contains all the information about it.*

**Example 1.3.** *For $\mathscr{C} = \mathrm{Top}$, the terminal object is again the singleton set and $\mathrm{Grp}\mathscr{C} = \mathrm{TopGrp}$.*

**Example 1.4.** *For $\mathscr{C} = \mathrm{Mani}^{\infty}$ the category of smooth manifolds, the terminal object is again the singleton set and $\mathrm{Grp}\mathscr{C} = \mathrm{LieGrp}$.*

**Example 1.5.** *For $\mathscr{C} = \mathrm{CAMani}^{\infty}$ the category of complex analytic smooth manifolds, the terminal object is again the singleton set and $\mathrm{Grp}\mathscr{C} = \mathrm{CALieGrp}$ the category of complex analytic Lie groups.*

**Example 1.6.** *For $\mathscr{C} = \mathrm{Sm}_k$ the smooth $k$-varieties for an algebraically closed field $k$, $\mathrm{Grp}\mathscr{C}$ is the algebraic groups over $k$.*

**Example 1.7.** *For $\mathscr{C} = \mathrm{Sch}/S$ the category of $S$-schemes, the terminal object is $S$ and$\mathrm{Grp}\mathscr{C}$ is the category of $S$-group schemes.*

**Definition 1.4.** *For an algebraically closed field $k$, An affine algebraic $k$-group is a group object in the category of smooth (affine) $k$-varieties.*

*An (affine) $S$-gropu scheme is a group object in the category of $\mathrm{Sch}/S$.*

**Definition 1.5.** *Let $\mathscr{C}$ be a category that admits finite co-products. A co-group object in $\mathscr{C}$ is a group object in $\mathscr{C}^{\mathrm{op}}$.*

**Example 1.8.** *Let $\mathbb{R}$ be a commutative unital ring and $\mathscr{C}$ the category of $R$-algebras which admits finite co-product. Then a co-group object in $\mathscr{C}$ is an $R$-Hopf algebra. Explicitly, for $A$ an $R$-algebra, we have*

$$\delta : A \to A \underset{R}{\otimes} A, \varepsilon : A \to R, \eta : A \to A$$

*which are called comultiplication, counit, and coinversion (or antipodal), respectively.*

**Remark 1.4.** *Recall that we have an equivalence of categories,*

$$(R\text{−Alg})^{\mathrm{op}} \overset{\mathrm{Spec}}{\to} \mathrm{AffSch}/R$$

*inducing an equivalence between $R$-Hopf algebra and affine $R$-group schemes.*

**Remark 1.5.** *Let $G$ be an affine $k$-variety and $\mathcal{O}_G$ be its ring of functions. Let $f \in \mathcal{O}_G, f : G \to k$, then*

$$f \circ m : G \times G \to k \in \mathcal{O}_{G \times G} \cong \mathcal{O}_G \underset{k}{\otimes} \mathcal{O}_G.$$

**Example 1.9.** *Let $R$ be a commutative unital ring, $\underline{\mathbb{G}_m}$ be the functor such that*

$$R\text{−Alg} \ni S \overset{\mathbb{G}_m}{\mapsto} (S^{\times}, \cdot) \in \mathrm{Grp}$$

*is the functor of points of the $R$-group schemes with Hopf-algebra,*

$$A := R[x, x^{-1}].$$

*Structure morphisms are given by*

$$A \ni x \overset{\delta}{\mapsto} x \otimes x \in A \otimes A, A \ni x \overset{\varepsilon}{\mapsto} 1_R \in R, x \overset{\eta}{\mapsto} x^{-1} \in A.$$

**Example 1.10.** *Let $R$ be a commutative unital ring, $\underline{\mathbb{G}_a}$ be the functor such that*

$$R\text{−Alg} \ni S \overset{\mathbb{G}_m}{\mapsto} (S, +) \in \text{Grp}$$

*is the functor of points of the affine $R$-group schemes with Hopf-algebra,*

$$A := R[x].$$

*Structure morphisms are given by*

$$A \ni x \overset{\delta}{\mapsto} x \otimes 1_A + 1_A \otimes x \in A \otimes A, A \ni x \overset{\varepsilon}{\mapsto} 0_R \in R, x \overset{\eta}{\mapsto} -x \in A.$$

**Example 1.11.** *Let $R$ be a commutative unital ring, $\underline{\mathrm{GL}_N}$ be the functor such that*

$$R\text{−Alg} \ni S \overset{\mathrm{GL}_N}{\mapsto} \mathrm{GL}_N(S) \in \text{Grp}$$

*is the functor of points of the affine $R$-group schemes with Hopf algebra,*

$$A = R[x_{11}, \cdots, x_{NN}, \det(x_{ij})^{-1}].$$

*The structure morphisms are,*

$$\begin{aligned}
\delta(x_{ij}) &= \left( \sum_{k=1}^{N} x_{ik} x_{kj} \right), \\
\varepsilon(x_{ij}) &= (\delta_{ij}), \\
\eta(x_{ij}) &= (\det(x_{ij}))^{-1} \mathrm{adj}(x_{ij}).
\end{aligned}$$

**Example 1.12.** $\underline{\mathrm{SL}_N}$ *is the functor sending*

$$R\text{−Alg} \ni S \overset{\mathrm{SL}_N}{\mapsto} \mathrm{SL}_N(S) \in \text{Grp}.$$

*Its Hopf algebra is*

$$A = R[x_{ij}]_{ij}/(\det(x_{ij}) - 1).$$

**Example 1.13.** $\underline{\mu_N}$ *is the functor sending*

$$R\text{−Alg} \ni S \overset{\mu_N}{\mapsto} \{x \in S^\times \mid x^N = 1\} \in \text{Grp}.$$

*Its Hopf algebra is*

$$A = R[x]/(x^N - 1)$$

*and the structure morphisms are the same as in $\underline{\mathbb{G}_m}$.*

**Example 1.14.** $\underline{\alpha_p}$ *for prime $p$ of the characteristic of $R$ is the functor sending*

$$R\text{−Alg} \ni S \overset{\alpha_N}{\mapsto} \{x \in S^\times \mid x^p = 0\} \in \text{Grp}.$$

*Its Hopf algebra is*

$$A = R[x]/(x^p)$$

*and the structure morphisms are the same as in $\underline{\mathbb{G}_a}$.*